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Building Condition Auditing (BCA)—Improving Auditability—Reducing Ambiguity

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Abstract: BCA methodically assesses the state of a building's deterioration to support Maintenance, Safety, Function, and Compliance purposes. Originally used to assist in identifying urgent repair requirements, it has evolved and become one of the most used tools for assessing a building's outstanding maintenance liability when a building is transacted or acquired. Nevertheless, current practices involve several conflicts; for example, high costs are associated with inspections, inconsistent building component registers, and ambiguity and consistency regarding reporting parameters, all of which lead to compounding errors that reduce reliability. To address these gaps, the current research, involving one hundred and eighteen (118) active facilities managers and asset inspectors, suggests the development of an extension of the deterioration scale (0–7) and methodologies to reduce errors and ambiguity. Furthermore, it suggests using weighted indices to focus on crucial building components, thus improving condition assessment. As was found, these tools improve the accuracy of BCA, facilitate better management of the asset's life cycle, and provide support in decision-making. This study adds consistency, limits subjectivity, and provides a framework applicable to different building types, assisting future management for sustainability. It, therefore, stands to serve the field by providing detailed and concise best practices for conducting condition audits on built assets.

Keywords: asset lifecycle assessment; BCA; building condition audit; building inspection

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1. Introduction

A building condition audit is commonly performed to address one of two operational objectives:

(a) An annual or routine condition assessment is performed as part of a strategic asset management plan to identify, document, and categorise an asset's current condition and trend its rate of deterioration over time. This is performed to develop a capital expenditure plan to return a building asset to its optimal operating condition.
(b) When a commercial, industrial, or institutional property is earmarked for sale and acquisition, a financial institution requests an independent building condition audit (technical due diligence report/TDD) as a precondition of sale or exchange to assess its current condition and identify potential maintenance liabilities.

The practice came into light during the middle of the 20th century because of the expansion of urban areas and poor structure standards. One of the reasons for developing systematic building maintenance is attributed to the development of urban areas because, in the past, the infrastructure deteriorated faster than it was being developed. At first, these assessments were ad hoc, with little more than cursory face-value evaluation, primarily to identify minor necessary works. However, by the 1970s, BCA had become common. Therefore, standards like the British Standard 8210, which called for frequent examination and thorough documentation to improve the authenticity of the assessment, were instituted

in the 1980s. The advancement of technology towards the closing years of the twentieth century changed BCA once again. Technologies such as Computerised Maintenance Management Systems (CMMS) provided BCA with adequate means of predictive maintenance to transform from the reactive type of company. Mentioned techniques such as infrared thermography, moisture meters, and ultrasonic testing provided the ability to see structural problems, making the assessments more efficacious [1]. In recent years, innovations such as Building Information Modelling (BIM), Geographic Information Systems (GIS), Internet of Things (IoT), and Digital Twins have shifted BCA practices, integrating the ability of real-time condition assessment, spatial data processing, and logical maintenance schedule preparation to develop more accurate BCA reports.

2. The Influence of Maintenance Practices on the BCA

Preventive maintenance, repair, and rehabilitation are all examples of tasks that fall under "maintenance". Building maintenance is a complex undertaking because of the enormous number of components in each building, each with maintenance requirements. A typical school building provides an excellent example of how difficult it is to manage built-environment assets. Foundations, exterior walls, interior doors, roof, boiler, transformer, etc., are only a few of the building's components. Figure 1 demonstrates what many people rarely see: the internal condenser vessel of an HVAC chiller exposed to poor water treatment, resulting in excessive scale buildup, which reduces heat/cooling transfer and increases power consumption. Similarly, in Figure 2, we can see the result of pipe internal scale built up over a long period, restricting water flow and increasing power consumption on the associated water pumps.



Figure 1. Internal Condenser Vessel.

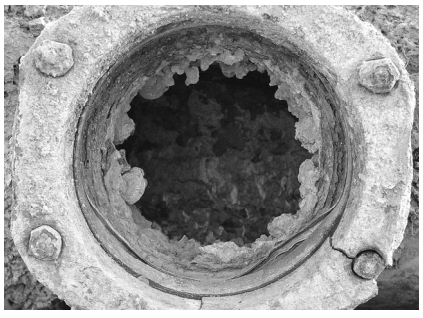


Figure 2. Internal wall water pipe.

In addition, a single building may include occupancy wear and tear demands on exposed building elements, meaning damage due to the building's operation, the amount and type of traffic, and even environmental exposure (e.g., wind, snow, setting sun direction, etc.). Facility managers across the sector are increasingly turning to this advanced preventative maintenance strategy. However, it has also resulted in the decentralisation and dispersion of critical lifecycle asset performance data, making strategic planning increasingly challenging to process and to make accurate determinations. The developing

nature of building technology and auditing tools, such as BIM and non-destructive testing (NDT) procedures, exacerbates this issue, necessitating ongoing professional development that impacts the building's operational budget. Furthermore, when operating budgets are constrained by competing market procurement processes, existing maintenance strategies are compromised and not updated to reflect the current assets' requirements as an emerging trend of continuing to use a corrective maintenance strategy instead of a preventative maintenance strategy later in a building asset lifecycle is becoming more frequent [2].

Additionally, when operating budgets are reduced, facilities managers often cannot access ongoing training budgets to stay updated with advancements in building condition modelling techniques and emerging computerisation applications, which leads to a reliance on outdated manual maintenance practices [3]. Facilities managers often need the requisite knowledge and training in building condition auditing, which ensures that building safety, functionality, and longevity can be optimised and strategically deployed in their daily operational routines. Many facilities managers are trained in general maintenance and operational procedures but frequently need specialised education in what is required to contribute to a thorough and effective building audit [4]. This training deficiency leads to insufficient evaluations, lack of historical detail, and incomplete disclosure of faults and errors, leading to oversight of crucial matters such as structural inadequacies, waterproof penetration issues, system malfunctions, or adherence to safety standards. Investing in specialised training programmes and certifications for facilities managers would address this knowledge deficiency, providing facilities managers with the essential foundational skills needed to understand what is required to perform and contribute to thorough and concise BCAs [5]. This lack of up-to-date knowledge and skills compromises the quality of BCAs and ultimately affects the maintenance and management of building assets [6].

3. Materials and Methods

The methodology describes the strategy for gathering, evaluating, and analysing the relevant data for this study [7]. The main goal is to guarantee the reliability and validity of the results while upholding strict academic standards. The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) diagram in Figure 3 has been used to undertake a systematic review and visualize the literature screening process deployed in this research assignment. The flowchart illustrates how researchers narrowed the initial search results to the final selection of journals and publications in multiple screening stages. The process began with identification, where databases were searched and records were collected. This is followed by screening, where duplicate records are removed and titles/abstracts are reviewed. During the eligibility assessment phase, full-text articles are evaluated against inclusion criteria. Finally, the included studies are determined and documented. Each stage is quantified, showing the number of studies retained or excluded. This standardized approach enhances reproducibility and allows readers to assess potential selection bias in the review process. The diagram's structured format ensures consistent reporting across different systematic reviews and meta-analyses, making it easier for researchers to compare methodologies and evaluate the thoroughness of literature searches.

This systematic review identified records, journals, and publications from 98 databases and sources, including Google Scholar, Scopus, Academia, ResearchGate, and Semantic Scholar, and involved publications from Frontiers, Buildings, Emerald, Elsevier, IEEE, Springer, Sage, and ScienceDirect. Title searches included Building Condition Audit, Building Condition Assessment, Building Condition Index, Facility Condition Index, AI and Building Assessments, and Different Analysis Methods for Building Assessments. The first screening level removed 9 publications due to duplications, resulting in a balance of 89 records. The second screening phase involved removing documents and publications that did not address the research theme or scope of building maintenance strategies. This resulted in 26 records being removed and a balance of 63 remaining journals and publications. Finally, 23 records were removed as they were not considered academically

appropriate or sufficiently peer-reviewed journals. The balance of reference documents, journals, and publications used in this study was 40 publications.

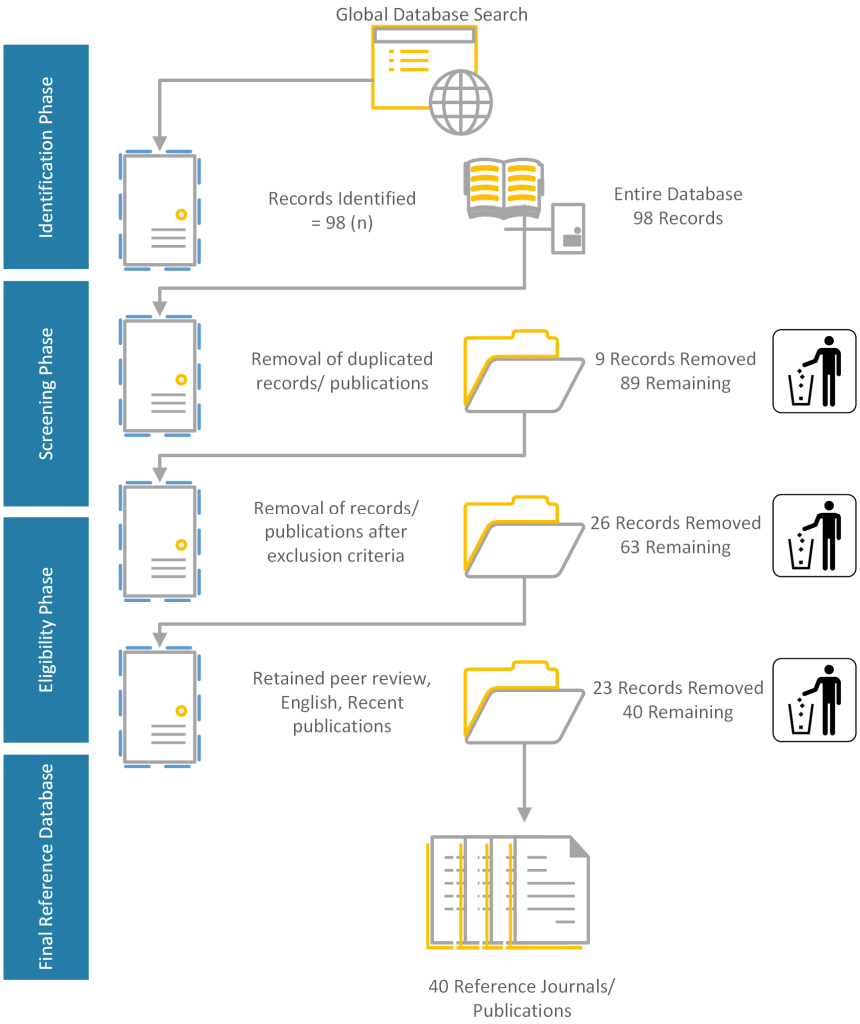


Figure 3. Prisma Model for Research Assignment.

The technical guides and reference materials in Table 1 were reviewed and evaluated to ensure that this research aligned with international, professional, and industry-recognised practices, which are essential when preparing a BCA or Technical Due Diligence Report.

Table 1. National Recognised Technical Reference Guidelines.

Country	Organisation	Reference Manual	URL
International	International Organisation for Standardisation	ISO 15686-5:2017 (en) Buildings and constructed assets—Service life planning	https://www.iso.org/standard/61148.html, accessed on 1 November 2024
United Kingdom	British Standards Institution	BS ISO 15686 Buildings and constructed assets. Service life planning Life-cycle costing	https://landingpage.bsigroup.com/LandingPage/Series?UPI=BS%20ISO%2015686, accessed on 1 November 2024
Australia	Royal Institute of Chartered Surveyors	Technical due diligence of commercial property, 1st edition	https://www.rics.org/profession-standards/rics-standards-and-guidance/sector-standards/real-estate-standards/technical-due-diligence-of-commercial-property, accessed on 1 November 2024
United States	ASTM	Building Facade Maintenance, Repair, and Inspection	https://www.astm.org/stp1444-eb.html, accessed on 1 November 2024

While Table 1 referenced recognised industry guidelines and standards used to perform a Building Condition Audit, Table 2 referenced academic journals and publications. This process ensured that the research being undertaken aligned with professional practices and the rigour of academic standards and peer-reviewed publications.

Table 2. Recent notable and peer-reviewed academic journal/ publications.

Journal	Author	Year
Analysis of Standardization and Guidelines for Facility Condition Assessments [8]	D. Hillestad; K. Sullivan; K. Hurtado; S. Ayer; J. Smithwick	2021
State of Practice for Facility Condition Assessment [9]	Pauline Karanja and Mayo, Glenda.	2016
Improving Equipment Maintenance—Switching from Corrective to Preventative Maintenance Strategies [2]	West, J.; Siddhpura, M.; Evangelista, A.; Haddad, A.	2024
Comprehensive Review and Comparative Analysis of Building Condition Assessment Models [10]	Begić, H. and Krstić, H.	2024
Facilities management: theory and practice [6]	Alexander, K.	2016
Asset maintenance in Australian commercial buildings [11]	West, J.; Siddhpura, M.; Evange-lista, A.; Haddad, A.	2024

Finally, to ensure that we captured the current market landscape, a survey was conducted involving one hundred and eighteen (118) respondents, including facility managers, asset managers, and field inspectors from commercial, government, and industrial properties in Canada, the USA, Australia, and Germany. Each response to the survey incorporated multiple-choice questions and questions with written answer options, including questions about the current BCA process, drawbacks of its proposed use, and ways for enhancing its efficiency [12]. The survey's goals were more straightforward, providing tangible information about practical concerns in the industry, such as cost, subjectivity, data integration, training, digitalisation, and emerging concerns.

The literature review and survey findings were analysed, and the data were sorted into nine identified themes: Technological Adoption, Standardization, and Assessment Consistency. The responses were aggregated to categorise them into new trends and repeated themes. The feedback loop was included so that respondents could check whether the derived themes were accurate and whether the findings were consistent with existing challenges in practice [13]. Further, the data analysis software used included Google Forms for survey collection and analysis and Microsoft Excel 365, version 2410 for data tabulation and theming. This allowed for the addition of transparency and probability while fulfilling the research's goal of establishing a set of recommendations for BCA improvements that could be replicated. Both methods enabled a plurality of approaches to BCA. They ensured the involvement of practical value derived from industry experience in refining an academic literature-based approach to developing a more effective auditing framework to meet industry needs while accounting for technological innovations [14].

4. Results

4.1. Building Condition Audit Background

Condition assessment specialists and organisations such as towns and government agencies would benefit from a well-structured, robust, and auditable decision-making process associated with BCA as it assists them in making the most cost-effective and sustainable funding and maintenance decisions possible [15]. Field inspection and condition assessment of building assets (such as commercial, industrial, retail, government, universities, etc.; see Figures 4 and 5) have been the primary focus of this research study, as this chapter examines several aspects of condition assessment, including asset hierarchy, evaluation processes, field inspection, and condition analysis, to give readers a complete picture of current state-of-the-art efforts [16].



Figure 4. Heritage Tertiary Education Building.



Figure 5. External Wall Inspection.

Subjectivity and the resulting lack of accuracy are significant roadblocks in creating a reliable condition assessment method. Condition assessments for buildings have traditionally been carried out by experts in the relevant building systems, such as structural and electrical. Nevertheless, the existing condition assessment procedure remains highly subjective and heavily reliant on the field inspectors' and assessors' expertise and education. With the current state of asset management and maintenance data being siloed from each other, a new comprehensive system of condition assessment is required [17].

The diverse nature of building components necessitates a broad range of expertise. Hence, inspectors must know about multiple domains, including structural engineering, electrical systems, Heating, Ventilation, Air Conditioning (HVAC), plumbing, and sustainability practices. However, it is rare for a single inspector to have a high degree of expertise in all these domains [5]. The absence of specialised knowledge leads to inaccurate or incomplete assessments, especially for intricate systems necessitating expert evaluation. Figure 6 exemplifies the typical categorisation of a typical building's assets categories involved in a building audit, providing high-level descriptors to the scope and detail associated with each audit category.

However, there are still many challenges that an asset inspector will face when having to perform a BCA or Technical Due Diligence Report, as exemplified in Table 3. The costs to undertake a thorough and detailed assessment remain very high. They are heavily dependent on the availability of a technically qualified and experienced workforce, significantly contributing to disparities in reporting and highly subjective assessment ratings [18]. Further, most facilities managers have little or no prior experience in the more sophisticated BCA tools used in today's systems analyses [19]. To address these challenges, this study aims to develop and suggest an improved 0–7 deterioration scale and adopt a weighted index that emphasises the most critical building component, giving a more accurate and transparent result. From the survey of and interviews with 118 facilities managers, and incorporating the latest technologies, this research proposes a refined BCA framework to assist in making sustainable asset management decisions on different infrastructures.

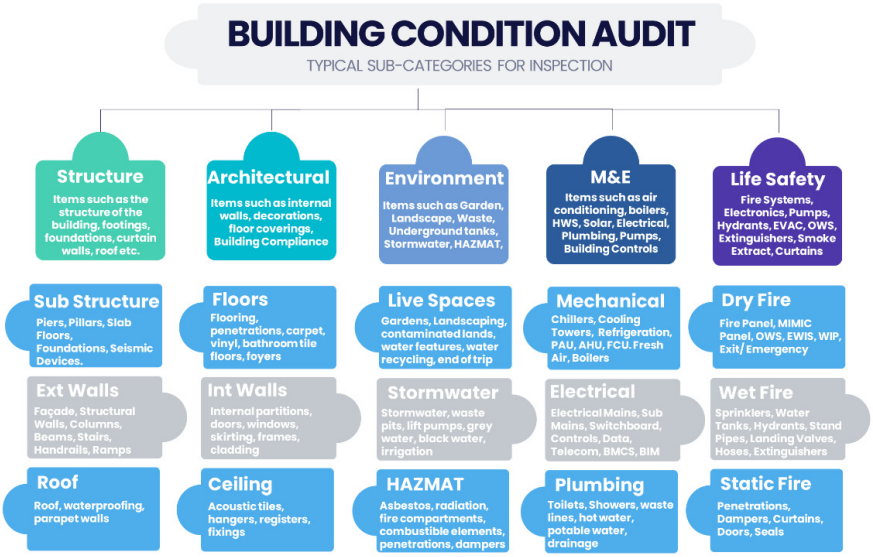


Figure 6. BCA Sub-Categories for building inspections.

Table 3. Challenges associated with Building Condition Audits.

Challenge	Responsible	Implication
Access to assets that are in difficult locations (roof, ceiling, wall cavities)	Asset Inspector	Incomplete and inaccurate Condition Assessment
Incomplete or inaccurate asset registers	Facilities Manager	Redundant and Disposed assets included in the assessment
Incomplete and inaccurate asset hierarchy	Facilities Manager	Quantity information regarding asset components cannot be computed
Incomplete or inaccurate building materials	Facilities Manager	Asset lifecycle longevity calculation will be inaccurate
Critical assets cannot be identified when equal ratings are applied	Asset Inspector	Smaller quantified assets (yet remaining critical) will be lost in the calculations compared with more significant volume asset components.
The deterioration scale is limited when using the traditional 0–5 rating, considering 5 = new and 1 = end of life.	Asset Inspector	One number on the traditional scale can result in multiple different deterioration conditions.
The defect recorded in an asset can vary depending on the asset inspector's knowledge and experience, and it is open to interpretation.	Asset Inspector	Significant variations in the deterioration record of the asset, resulting in ambiguity, confusion, and discredit

The fundamental requirement for establishing a BCA is to

(a) create a detailed building asset register that includes all the building elements, services, and infrastructure and quantify each component based on size, volume, composition, and quantity.
(b) Establishing a hierarchical structure for the asset register is crucial. This includes identifying each level and room within the building and assigning and marking asset inventory codes in strategic locations.
(c) Assign an asset key (asset ID code) for each asset component that reflects the building, area, floor, room, and component.
(d) assign the theoretical lifecycle age of the different building components and cross-reference them against a recognised industry standard for asset deterioration, like the Australian Chartered Surveyors Guide (ACSG) [20],
(e) determine each component's theoretical remaining lifecycle age.

Once the above phases have been completed, documented, and assigned entry into a detailed asset register, an inspector can commence rating each building component [9,21].

In recent years, a trend towards sustainability and energy efficiency has impacted the scope of BCAs. Contemporary audits frequently encompass assessments of a building's environmental efficacy, including energy consumption, carbon emissions, and adherence to sustainability criteria [22]. This expanded emphasis signifies the increasing acknowledgment of the significance of sustainable construction methods in mitigating environmental effects and enhancing building efficacy. The paramount purpose of asset management is condition assessment, which acts as the cornerstone for all subsequent operations, including the decision to repair or replace an asset. Condition assessment means the repair, renewal, or replacement of an organisation's capital assets to ensure they can continue supporting the mission or activities they have been assigned. It remains a strategic maintenance decision-making process when determining the whole life of property asset management [23]. This change indicates increasing understanding from the as-built building sector endorsement of these valued matrixes, highlighting emerging environmental concerns and increased assets' durability. In addition, the matrixes emphasise' the industry's interests in ensuring that when buildings are transacted or exchanged, they align with global interests and will achieve market perceptions and interests in securing and retaining future tenants [11].

4.2. Building Condition Audit Formulas/Methodologies

When an organisation considers performing a building condition audit evaluation of their building assets components, three different options are available:

(a) Distressed Condition Audit
(b) Direct Condition Rating Audit; and
(c) The Health Index Audit

For a distressed condition audit to be accurate, it must individually record all the flaws in each building component being evaluated independently. A comprehensive systematic approach for a trained field inspector is crucial for effectively detecting, determining, and assessing the condition of the building. Table 4 exemplifies a simple distressed condition audit incorporating a criticality factor.

Table 4. A Distressed Condition Audit Example.

Building Component	Condition Rating (CR)	Criticality Factor (CF)	Component Score (CS) (CR × CF)	Distress Factor (DF) (= 25 − CS/25)
Foundations	5	5	25	0.0
Exterior Walls	4	5	20	0.2
Windows	3	3	9	0.64
Roofing	4	3	12	0.52
Interior Walls	3	2	6	0.76
Interior Ceiling	4	2	8	0.68
Flooring	2	3	6	0.76
Finishes	2	1	3	0.88
HVAC&R	3	3	9	0.64
Electrical	3	3	9	0.64
Plumbing	4	3	12	0.52
Life Safety	3	5	15	0.4

As demonstrated in Table 4 (Column Distressed Factor Column), the closer the index figure reaches 1.00, the higher the distressed factor or increased need for maintenance intervention, maintenance liability, or area of risk in a building acquisition process.

Conversely, a direct-condition rating is less accurate but considerably more expedient than the duress condition audit approach, as each element is visually examined and assessed against a predefined scale with a set degradation criterion (Table 5). Both a duress condition audit and a direct-condition rating survey can assess the health of a single building's component at any given time relative to the building's total lifecycle [24].

Table 5. Building Condition Assessment performed in 2024.

Building Component	Built or Refurbish	Theoretical Life (TL)	Deterioration Years	Remaining Life (RSL%)	PCS	DCR
Foundations	2010	75	14	0.85 (85%)	5	4.25
Ext. Walls	2010	50	14	0.72 (72%)	4	2.88
Windows	2010	35	14	0.60 (60%)	3	1.80
Roofing	2010	40	14	0.65 (65%)	4	2.60
Int. Walls	2010	25	14	0.44 (44%)	3	1.77
Int. Ceiling	2010	25	14	0.44 (44%)	4	1.76
Flooring	2023	15	1	0.93 (93%)	5	4.65
Finishes	2020	7	3	0.57 (57%)	4	2.28
HVAC&R	2010	25	14	0.44 (44%)	3	1.32
Electrical	2010	30	14	0.53 (53%)	3	1.59
Plumbing	2010	35	14	0.60 (60%)	4	2.40
Life Safety	2022	15	2	0.86 (86%)	5	4.30

PCS = Physical Condition Score (actual condition based on an assessment). DCR = Direct Condition Rating.

As demonstrated in Table 5, the higher the number, or closer to the rating of 5, the newer the asset component is and the less of a maintenance liability it represents. However, over time and use, the lower the numbers get and the closer to zero, the higher the risk factor and necessity for maintenance intervention.

A different audit methodology is the Health Index Audit (HI) (Equation (1)), which evaluates the condition of individual building components based on their physical state, performance, and remaining economic life. Each component is scored on a scale (e.g., 0 to 100), and the overall building condition index is derived by aggregating these scores.

Example:

Roof Rating	= 60 out of 100
Superstructure Rating	= 60 out of 100
HVAC Rating	= 20 out of 100
Plumbing Rating	= 50 out of 100

$$HI = \frac{Roof\ Rating + Superstructure\ Rating + HVAC\ Rating + Plumbing\ rating)}{\text{Number of Componant Catagories}}$$

$$HI = \frac{(60 + 60 + 20 + 50)}{4} \quad (1)$$

$$\mathbf{HI = 47.5}$$

Whatever BCA survey protocol is utilised, it is a reliable and repeatable method of collecting building condition deterioration data. In all three methodologies, the essential part of assessing the building's condition is breaking it down into its primary components of a hierarchical structure tree, as these components are further broken down into separate categories and subcategories. Different disciplines or systems, such as electrical and mechanical, can be subdivided into more specific components, such as interior doors, exterior doors, windows, and ceilings. Similar qualities or inspection requirements may dictate the placement of components in a particular branch of the hierarchy tree [25]. The

building condition survey of a building's systems and components is based on evaluating its current condition. The more time passes between inspections (for example, annually), the more thorough the assessment will be, as regular condition assessments make the assessment process considerably more straightforward, as demonstrated in Figure 7. When it comes to the frequency of condition evaluations, the cost of the inspection is a significant consideration. During the field inspection, gathering adequate detailed information is crucial. If the data collected are excessively detailed, it may be unnecessary and a waste of time—conversely, insufficient information results in wasted time and resources [26].

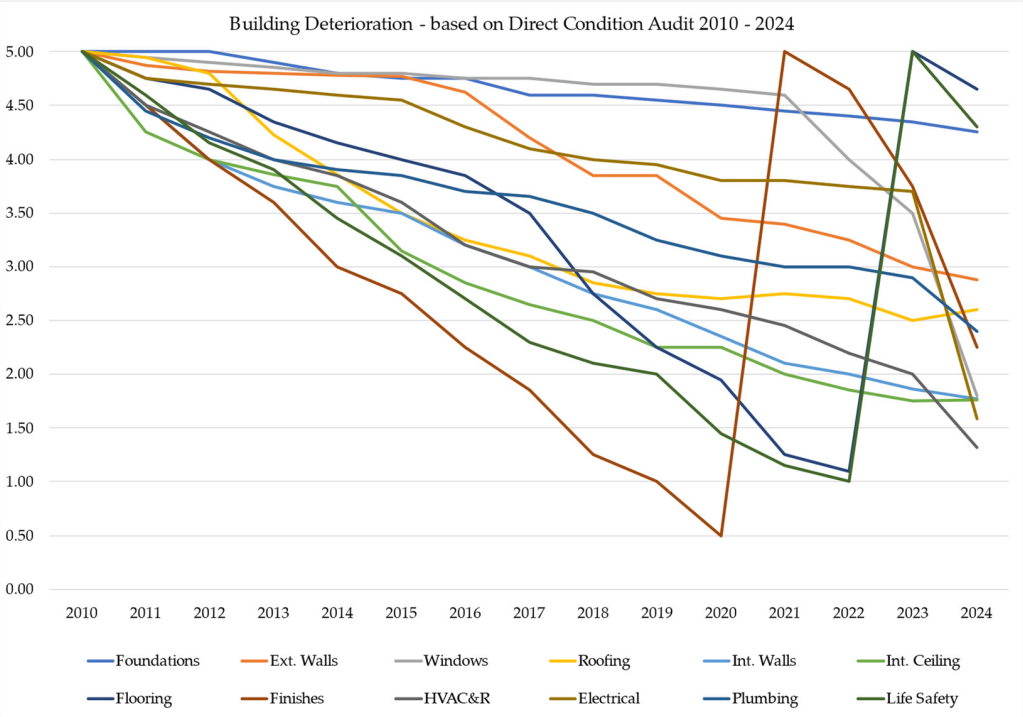


Figure 7. Building Deterioration Trend (Based on Direct Condition Audit 2010–2024.

The effectiveness of BCAs significantly depends on the inspectors' consistency and technical competency. Inconsistencies in assessments and variations in inspector expertise can result in unreliable audit outcomes, fluctuating Building Condition Index numbers, and challenges in developing algorithms to support digitisation, automation, and reporting. For the inspection process to succeed, data must be collected to assess and compute performance or evaluate the relevant state and deterioration. Inspections should be conducted as regularly, accurately, and objectively as possible. Establishing checklists and defect lists for inspection is critical for standardising the process and ensuring consistent reporting and deterioration tracking, whether printed or electronic. Another approach explored by research academics is automating the examination process with robots, satellite imagery, drones, and intelligent sensors [27]. Even though a component's condition can be measured in terms of its deterioration severity, some analysis is still necessary to convert these measurements into a recognised condition rating or value [28]. The condition of a component can be evaluated at any level of the asset hierarchy using condition aggregation.

When deciding between direct rating or distress survey, the evaluation goal must be considered before deciding which method to use. The direct-condition rating approach is sufficient if the component's condition is exciting and will contribute to a trend of deterioration over time. However, the distress survey approach should be used if the goal is to identify current issues, specific in time, with no intent or desire to trend or track deterioration. Regardless of which methodology is used, the most debated topic remains the accuracy or subjectivity of the field inspection determinations, which depend highly on the inspector's thoroughness (time), technical knowledge, evaluation process,

and accessibility. Existing systems require a skilled inspector to be present throughout the inspection process to evaluate the asset's condition based on all relevant criteria [8].

5. Research Limitations

The scope of this assignment involved conducting a Building Condition Audit (BCA) and developing a reliable Building Condition Index (BCI). The authors acknowledge that industry practices include a Facility Condition Index (FCI), a financial reporting metric often used in Technical Due Diligence (TDD) Reporting and Audits. However, the FCI has been deliberately excluded from this assignment to ensure focused attention on the current challenges associated with Building Condition Auditing (BCA) and the development of the Building Condition Index (BCI). The analysis excludes various software applications, including Building Management Computer Systems (BMCS), Computerised Maintenance Management Systems (CMMS), and Building Information Modelling (BIM) software, as well as other specialised government software, such as Software Asset Management (SAM) and the Precipitation-Runoff Modelling System (PRMS) [29]. These applications are built on the core foundation of Building Condition Auditing (BCA) and Building Condition Index (BCI) methodology. Without clearly defining the issues, challenges, and complications at the foundation level, it would only complicate and weaken any complimentary software or application that relied on these principles. While this assignment offers insight into the factors that drive the decision-making process associated with Building Condition Audits, it was based on one hundred and eighteen buildings in Australia, Canada, the US, and Germany. All efforts have been made to develop a dataset based on a diverse portfolio of different buildings (commercial, industrial, retail, education, institutional, etc.); however, the outcomes and findings are limited to this dataset. Specifically, additional or altered perspectives might have been attained if the research scope was extended to include different commercial buildings, a broader range of operating and maintenance strategies, and different geographical regions and building occupant types. Further limitations arise due to corporate culture, interpersonal dynamics, external influences, technological advancements, economic shifts, and changes in corporate objectives, which were not explored in this research.

6. Ethical Considerations

Ethical considerations in modern building management strategies are critical to ensuring the safety, reliability, and overall health and well-being of all stakeholders, including tenants, service providers, contractors, visitors, and people in the broader community [30]. The survey and interviews with service providers and facilities managers revealed a strong sense of professional responsibility, diligence, and desire to ensure equipment functionality, safety, and operability. Their responses to multiple questions underscored the facilities managers' vigilance towards occupational health and safety. The stakeholders also demonstrated a strong focus on sustainability targets and objectives ranging from green energy solutions to waste reduction and minimising carbon emissions impacting society, suggesting that they were committed to ensuring the long-term commitment to providing and maintaining a high-value building asset [31].

7. Discussion

Based on the research and thematic results from surveys and consultation sessions with facility managers, service providers, and asset inspectors, a summary sets the scene for this paper's originality. Significant advancements in this field would be achieved when a structured, creditable, and consistent reporting asset assessment and deterioration methodology can be replicated and duplicated by any field asset auditor regardless of skill, expertise, training, or experience. The following outcomes have been summarised based on the thematical responses from the qualitative survey and thematic surveys performed. Below is a detailed analysis of the findings from this research assignment, identified, correlated, and categorised into the ten most prominent themes.

7.1. Subjectivity in Assessments

One of the primary challenges in BCAs is the inherent subjectivity in assessments. Despite following standardised checklists, inspectors have different interpretations of building conditions. This subjectivity leads to inconsistencies in identifying defects and their severity, affecting audit reports' reliability [10]. This highlights the need for more objective evaluation methods to mitigate subjectivity and improve the consistency of assessments.

7.2. The Lack of Standardised Training

The absence of standardised training programs for building inspectors exacerbates inconsistency. Training programs and certifications vary widely, resulting in differing levels of knowledge and skills among inspectors [18]. Ensuring that all inspectors have the requisite expertise to perform thorough and accurate audits is only possible with a universally accepted training standard.

7.3. Variability in Reporting Practices

Reporting practices also need more standardisation, their absence contributing to inconsistencies. Inspectors use various formats and terminologies, making comparing and consolidating audit reports difficult. This variability hampers tracking building conditions over time and prioritising maintenance tasks effectively [4].

7.4. Limited Rating Structured Logic

Qualitative survey results revealed a strong theme indicating that rating building components on a 0–5 scale was limiting and restrictive. A rating of five was typically assigned for new buildings and structures, while buildings or structures at the end of their economic life were rated as one or zero. This left asset inspectors with only three rating categories—two, three, and four—to utilise.

7.5. Dilution of Critical Components

A theme highlighted in the qualitative survey was the complexity of different building elements and how critical defects or faults were identified. The invasion of termites in the wooden structure exemplified this theme, as the damage was considered severe, and the risk rated extremely high. However, unless a significant portion of the wooden structure—such as studs, noggins, and top and bottom plates—were destroyed, the sheer volume of surveyed wooden elements would dilute the defect's impact on the overall rating of this building component. When amalgamated with roofing components, services, floor, foundation, etc., the severity of the defect becomes diluted. It runs a high risk of not being escalated to a level when an immediate intervention or repair solution is required to preserve the remaining building assets.

7.6. Poor Building Structure/Missing Components

Following establishing a thorough and detailed asset register, each building needs to be separated according to the floor, level, room, and the associated components that comprise these areas. This would involve 4 × walls, a ceiling, a floor and all the building services. However, consideration also needs to be given to the structural elements and establishing a hierarchical structure that will enable the rating of the surfaces and finishes to be separate from the structure. Furthermore, it is necessary to appreciate that a building has an external wall and internal walls, along with an external roof and an internal ceiling, which require assessment and will deteriorate at different speeds and rates.

7.7. Inconsistent Building Component Lifecycle

Determining the length of remaining economic life concerning a building component is critical, and the database must be established with a clear and consistent standard for determining the building components' life expectancy. For example, an external masonry brick wall will last between 40 and 75 years, while profile steel cladding lasts only 15 to

20 years [27]. Adopting an industry-wide accepted and recognised component lifecycle duration will further assist in comparing different buildings, the rate at which they deteriorate, and the impact that building material quality plays on the building's economics.

8. Conclusions

The thematic survey and literature review results suggest that BCA should become more standardised and less opaque. Some of the participants' concerns were the irregularities that stemmed from differences in the auditors' skills and knowledge, which affected the level of audit reliability and subjectiveness of the inspections [32]. The extended deterioration scale from the traditional (0–5) to an improved (0–7) scale proposed in this research was received well, as it introduced more differentiated asset conditions in the mid-range area, as demonstrated in Figure 8 [33].

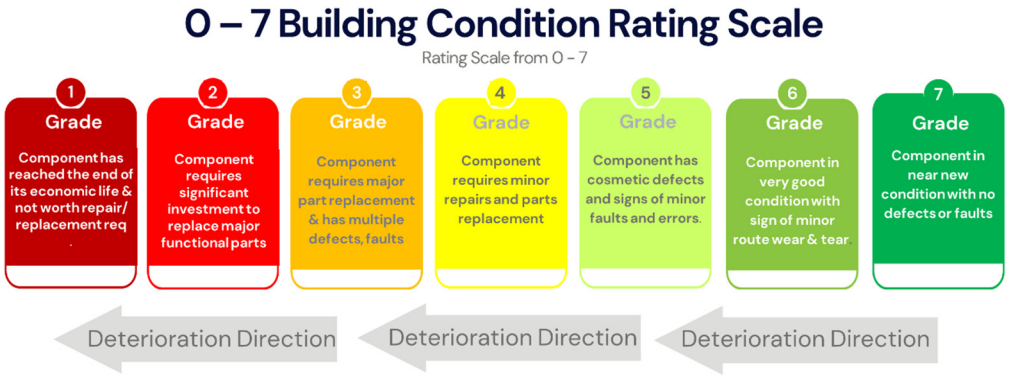


Figure 8. Standard baseline 0–7 Deterioration Scale.

By following the prompts and condition rating descriptors or guides noted in the deterioration scale examples in Figure 8, an asset inspector can reference the deterioration observations for different building components and develop a corresponding deterioration rating specific to the building material being assessed, as exemplified in Figure 9.

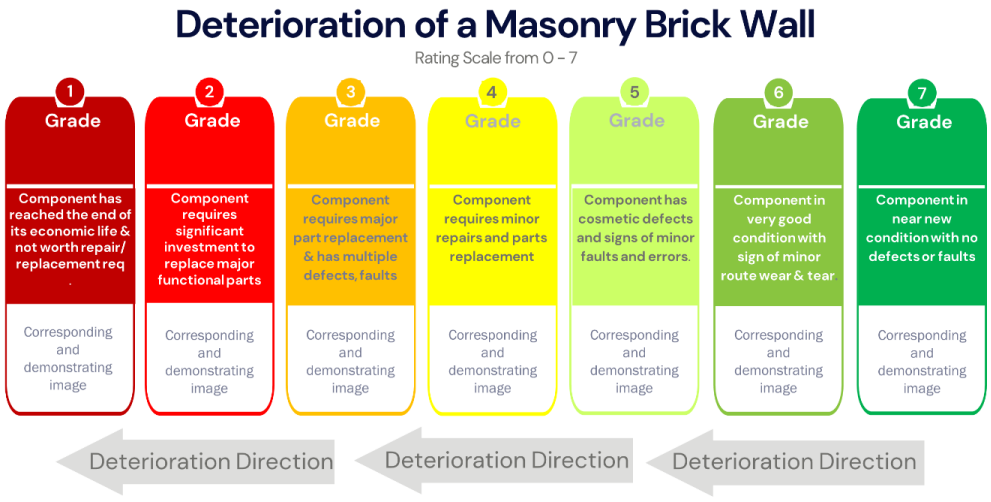


Figure 9. A 0–7 Deterioration Scale specific for Masonry Brick Walls.

Establishing structured asset identification strings is essential for accurately classifying a building's components and linking them to the corresponding room, floor, area, and building. Establishing asset identifiers corresponding to a parent asset identifier supports linking and tying the building components together to facilitate the accurate condition assessment and assignment of deterioration curves. Furthermore, it establishes the groundwork for accurately accounting for the building's assets, Capital Expenditure (CAPEX)

upgrades, and asset replacement activities. Figure 10 is a graphical representation of a well-structured hierarchical diagram of parental assets.

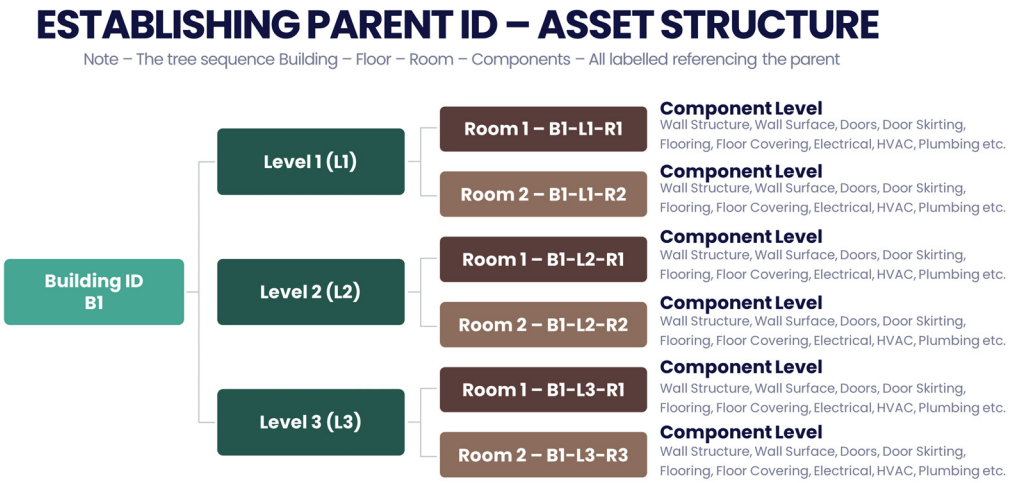


Figure 10. Asset Component Hierarchy Chart.

As buildings become increasingly complex and our pursuit for perfection continues, the expansion of helpful software becomes increasingly essential. However, as we capture more building components, a significant but subtle defect may be diluted or overlooked among the other components unless a discrimination matrix is used. By employing a discrimination matrix, faults or errors can be escalated appropriately, ensuring that critical issues are mathematically highlighted and brought to attention. Moreover, designing the discrimination matrix and the structured asset identification system in BCA shows that each context should be evaluated, and repairs and maintenance priorities should be prioritised by severity. Before the additional attributes are discussed, it is relevant to note that they are in contrast with the traditional BCAs' weaknesses, such as data silos and reporting variability, which these additions make less of an issue, allowing the facilities managers to address the identified risks [34] better.

The occurrence is beneficial because the matrix enables defects that could otherwise be concealed by low-risk data points to be as they should [35]. This improvement responds to the industry's question of ensuring that those issues demanding attention are noticed. Moreover, the survey of one hundred and eighteen participants' feedback also showed that integrating technology such as BIM and GIS with standard condition rating templates will facilitate data integration and reporting [36]. Altogether, these discoveries point to utilising extended metrics with BCA technological integration to improve precision and the tool's applicability for various building types and requirements for different stakeholders. Last, the findings underline that technology management, in general, and BIM and GIS, in particular, extend beyond the optimisation enabler and serve as a transparency enhancer, practical information and communication flow facilitator and enabler of lifecycle asset management. Future research may examine how these improvements have been made in specific cases to analyse the advantages of the implementation process's cost, time, and accuracy.

By incorporating technology such as BIM and GIS into this research, this study moves towards change within the facility management industry by adopting real-time data to track building conditions and enhancing stakeholder communication [37,38]. All these tools collectively have a role in improving asset management's sustainability as they provide tools to support decisions in repair, replacement, and asset life cycle. The potential of this enhanced framework might be tested and demonstrated in live working environments; therefore, future work could develop ways of using this improvement in actual settings to show its effectiveness and flexibility. In trying to fill the gaps between specific conventional inspection methods and practices and data enabler approaches, this study may create a

knowledge base for exploring and advancing the BCA practices suitable for sustainable and efficient asset management.

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Abbreviations

BCA	Building Condition Audit (or Assessment)
BCI	Building Condition Index
BIM	Building Information Modelling
CMMS	Computerised Maintenance Management Systems
DCR	Direct Condition Rating
GIS	Geographic Information Systems
HVAC	Heating, Ventilation, Air Conditioning
IoT	Internet of Things
M&E	Mechanical and Electrical
PCS	Physical Condition Score

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